

Customer Churn EDA Report

1. Introduction

Customer churn refers to the phenomenon where customers discontinue using a company's products or services. Churn analysis is important because acquiring new customers is generally more expensive than retaining existing ones. Understanding the factors that influence customer churn helps organizations improve customer satisfaction, increase retention rates, and enhance overall business performance.

This report presents an exploratory data analysis of a customer churn dataset to identify patterns and factors associated with customer attrition.

2. Dataset Details

The analysis was performed using the **Telco Customer Churn Dataset** obtained from **Kaggle**.

Dataset Information

- **Dataset Source:** Kaggle
- **Dataset Name:** Telco Customer Churn Dataset
- **Number of Rows:** 7,043 customer records
- **Number of Columns:** 21 features
- **Target Variable:** Churn

Numerical Features

- SeniorCitizen
- Tenure
- MonthlyCharges
- TotalCharges

Categorical Features

- Gender
- Partner
- Dependents
- PhoneService
- InternetService
- Contract
- PaymentMethod
- Churn and other service-related variables

Each row in the dataset represents information corresponding to a single customer.

3. Data Cleaning Summary

Several preprocessing steps were performed to ensure the quality and consistency of the data.

Missing Values

Blank values were identified in the **TotalCharges** column. These values were converted into null values and replaced using the median value of the column to preserve the distribution of the data.

Duplicate Records

Duplicate rows were checked and removed to prevent redundant observations from affecting the analysis.

Data Type Corrections

The **TotalCharges** column was initially stored as an object type due to the presence of blank spaces. It was converted into a numerical format to facilitate statistical analysis.

Unnecessary Columns

The **customerID** column was removed because it does not contribute to customer churn analysis and serves only as a unique identifier.

Inconsistent Values

Extra spaces and formatting inconsistencies in categorical variables were removed to maintain uniformity.

Outlier Detection

Outliers in numerical features were examined using boxplots and the Interquartile Range (IQR) method. Although some extreme values were observed, they represented genuine customer records and were retained for analysis.

4. Exploratory Data Analysis

4.1 Univariate Analysis

Univariate analysis was conducted to understand the distribution of individual variables.

Churn Count Plot

The count plot revealed that the number of customers who did not churn was significantly higher than the number of customers who churned.

Observation:

Most customers remained with the company, while only a smaller portion discontinued the service.

Churn Percentage Pie Chart

The pie chart showed the proportion of churned and non-churned customers.

Observation:

Approximately 26–27% of customers had churned, whereas nearly 73–74% continued using the services.

Histogram of Tenure

The histogram illustrated the distribution of customer tenure.

Observation:

A large number of customers had shorter tenures, while comparatively fewer customers remained with the company for longer periods.

Boxplot of Monthly Charges

The boxplot was used to detect unusual values in monthly charges.

Observation:

Some outliers were observed, indicating customers with exceptionally high monthly bills.

Gender Distribution Bar Chart

The bar chart compared the number of male and female customers.

Observation:

The dataset contained nearly equal numbers of male and female customers, indicating balanced gender representation.

4.2 Bivariate Analysis

Bivariate analysis was performed to study the relationship between customer attributes and churn.

Contract Type vs Churn

Customers with month-to-month contracts exhibited the highest churn rates.

Observation:

Long-term contracts were associated with lower churn, indicating stronger customer retention.

Internet Service vs Churn

Fiber optic customers experienced relatively higher churn rates than DSL customers.

Observation:

Service quality or pricing factors may influence churn among fiber optic users.

Monthly Charges vs Total Charges Scatter Plot

The scatter plot examined the relationship between monthly charges and total charges.

Observation:

Customers with higher monthly charges tended to have higher total charges and showed relatively higher churn behavior.

4.3 Numerical Analysis

Descriptive statistics including mean, median, minimum, maximum, and standard deviation were calculated for numerical features.

Observation:

- Monthly charges exhibited considerable variation among customers.
 - Total charges showed a wide range because they accumulated over customer tenure.
 - Standard deviation values indicated moderate variability in the numerical variables.
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4.4 Categorical Analysis

Value counts and category-wise churn comparisons were performed.

Observation:

- Month-to-month contracts were the most common.
 - Electronic check was one of the most frequently used payment methods.
 - Non-churn customers outnumbered churn customers.
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4.5 Correlation Analysis

Correlation analysis was conducted to study relationships among numerical variables.

Observation:

- Tenure exhibited a negative relationship with churn, indicating that long-term customers are less likely to leave.
 - Monthly charges showed a positive association with churn.
 - Total charges demonstrated a moderate relationship with churn.
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4.6 Outlier Analysis

Boxplots and the Interquartile Range (IQR) method were used to identify extreme observations.

Observation:

- Outliers were present in MonthlyCharges and TotalCharges.
 - These extreme values corresponded to valid customer records and were retained.
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5. Key Findings

The following insights were obtained from the analysis:

1. Approximately 26–27% of customers have churned.
2. Customers with month-to-month contracts exhibit the highest churn rates.
3. Customers with shorter tenure are more likely to leave the company.
4. Higher monthly charges are associated with increased churn.
5. Fiber optic customers show relatively higher churn compared to DSL customers.
6. Contract type, tenure, and monthly charges are the most influential factors affecting churn.

6. Business Recommendations

Based on the findings, the following strategies are recommended:

- Encourage customers to adopt long-term contracts through attractive discounts and incentives.
- Implement loyalty programs to retain customers with shorter tenure.
- Provide customized pricing plans for customers with high monthly charges.
- Improve the quality of fiber optic services to reduce dissatisfaction.
- Strengthen customer support and engagement programs.
- Use predictive analytics to identify customers who are likely to churn and proactively offer retention strategies.

7. Conclusion

This exploratory data analysis provided valuable insights into customer behavior and the factors contributing to churn. The results indicate that contract type, tenure, and monthly charges are the primary determinants of customer attrition. By focusing on customer retention strategies and improving service quality, organizations can reduce churn rates and enhance long-term customer relationships. Effective churn management ultimately contributes to increased profitability and sustainable business growth.